Problem-Driven Educational Technology for Ghanaian Higher Education: An Offline-First Learning Management Engine and UTAUT Implementation Protocol

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ABSTRACT

Higher educational institutions across Ghana face chronic infrastructural constraints that undermine digital learning initiatives following rapid post-pandemic enrollment growth. This paper investigates the top five pedagogical and infrastructural obstacles facing Ghanaian university students and designs a zero-rated, offline-first learning content engine. We analyzed empirical data from the Ghana Tertiary Education Commission (GTEC) and regional university repositories, applying formal scoring criteria across scale of impact, urgency, empirical evidence, and technical tractability. This research targets the five primary educational technology bottlenecks in Ghana: mobile internet data costs, frequent power load shedding, faculty LMS adoption resistance, indigenous curriculum scarcity, and virtual laboratory deficits. As a tangible open-source artifact, we designed, built, and tested an offline-first learning management engine featuring zero-rated micro-content packaging and asynchronous synchronization in /prototypes/edtech/.

Keywords: Educational Technology, Offline-First LMS, UTAUT Protocol, Ghanaian Higher Education, Digital Divide, Load Shedding Mitigation

PROBLEM-TO-SOLUTION SUMMARY & OPEN ARTIFACT

This publication presents an empirical pathway connecting established computational and engineering theory to physical, field-validated artifacts addressing five ranked national problems in Ghana. A runnable, tested open-source prototype has been authored and published under an MIT license in the repository directory /prototypes/edtech/ with automated unit test suites.

Canonical Web Reader: https://bernardfiagbenu.vercel.app/research/edtech